Knowledge Engineering for the Policy-Aware Personalized Opportunity (PAPO) Framework: A Companion Ontology and Flowchart, With Heatwave Response as a Worked Example

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Companion to: Wu, Min. *PAPO: A Policy-Aware Personalized Opportunity Framework for Heatwave Public Health Interventions.* SSRN, 2026 [1]; and Wu, M. *CMDDS — PAPO Heatwave Framework, v1.2.* Zenodo, 2026 [2].

Abstract

This is a companion paper to the framework's founding description [1] and its operational specification [2]; it assumes the reader already has that context and does not re-introduce the Policy-Aware Personalized Opportunity (PAPO) framework. Its sole focus is a knowledge-engineering development undertaken since those publications: PAPO formalized as a two-layer, OWL-style ontology in YAML — a hazard-agnostic core and a heatwave-specific extension — together with a single flowchart rendering the framework's core translation logic as a diagram. Both artifacts are new additions to the framework's open GitHub repository. We report the knowledge-engineering process that produced them, the resulting structure, and why we consider the flowchart — not the formal axioms behind it — the artifact most likely to make PAPO's logic legible to readers who will not work through a full specification. Both artifacts are openly available; this paper is their plain-language companion.

Keywords: knowledge engineering, ontology, middle-range theory, public health informatics, heatwave, policy translation, conceptual modeling

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1. Introduction and Scope

This is a companion paper to the framework's founding description [1] and its operational specification, the Conceptual Model Design Documentation for Simulation (CMDDS) [2]. Readers unfamiliar with the Policy-Aware Personalized Opportunity (PAPO) framework — a middle-range theory holding that personalized, feasible action outperforms generic alerts during time-critical public health events, first operationalized for heatwave response — should start there. This paper does not re-introduce PAPO. It reports, and makes legible, a knowledge-engineering development undertaken since that publication.

The CMDDS is implementation-agnostic by design and explicitly invites forks: city-specific adaptations, alternative simulation paradigms, comparative studies. Six months after publication, it has received attention but no forks. We do not think this reflects a flaw in the underlying theory.

We think it reflects a gap between *specifying* a system precisely and *presenting* it legibly. CMDDS Step 3 describes what it calls the framework's “non-negotiable logic core” — the translation from policy to feasible opportunity — carefully, across several pages of entities, attributes, and rules. Precise, but slow to absorb, and slow-to-absorb is often equivalent to unused.

This paper reports a knowledge-engineering pass undertaken to close that gap: a formal, two-layer ontology of PAPO's concepts and relationships, and a single flowchart rendering its core logic as a diagram rather than a specification document. Both are new since [1] and [2], and both are openly available in the `knowledge-engineering/` directory of the framework's GitHub repository (https://github.com/Min-PH/PAPO-Heatwave-AI). We describe the process (Section 2), the resulting structure (Sections 3–4), and why we believe the diagram — not the ontology behind it — is the artifact most likely to actually get read (Section 5).

2. Why Knowledge Engineering, and Why Now

A prose specification and a formal ontology answer different questions. Prose is good at explaining *why* a design choice was made. It is weaker at answering, unambiguously, *what exactly is being claimed* — is a “Resource” the same kind of thing as a “Policy”? Can an “Underserved Population” also be a “Vulnerable Population”? Is null output a failure state, or a required outcome? A reader has to infer these boundaries from context; two readers can reasonably infer differently.

An ontology forces these boundaries into the open. Classes must be explicitly related or explicitly separated (a disjointness axiom is a declaration that two things can never be confused); required conditions must be stated as such (e.g., every deployed policy must have an approved equity metric, not merely encouraged to have one); and structural distinctions the prose draws informally — tactical versus strategic feedback, activation versus escalation triggers — become distinct, checkable classes. None of this changes what PAPO claims. It changes how quickly and how confidently a new reader — or a future collaborator's software — can determine what PAPO claims.

A concrete example: the CMDDS states that when no feasible opportunity exists, the system should output null rather than force a poor match, and separately calls this a “required architectural element” rather than an edge case. A prose reader can hold both statements without necessarily registering their relationship. Formalizing the same claim as an equivalence axiom — a null opportunity simply *is* an opportunity with its null-output flag set true, not a different kind of thing — makes the relationship unambiguous and, in the same step, forces a builder's software to actually produce that state rather than merely handle it as an unplanned failure mode.

There is also a practical, forward-looking reason. A framework intended for reuse across hazards, as PAPO's founding paper explicitly claims heatwave application is a first instance of, benefits from a form that separates what is theory-level from what is hazard-specific. A prose document tends to blur that line, because the illustrative examples (cooling centers, heat alerts) sit inside the same sentences as the general claims. A layered ontology enforces the separation structurally.

3. Method: A Two-Layer Modeling Strategy

We formalized PAPO as two files: a hazard-agnostic **core** ontology and a heatwave-specific **extension** that imports it. The split was not decided in advance; it emerged from repeatedly asking, of every concept in the CMDDS, “would this survive being applied to a different hazard?” Concepts that would (a Population Segment's underlying vulnerability logic, the requirement that scarce resources be allocated by a governance-approved rule, the five recognized ways an opportunity can fail to materialize) went into the core. Concepts that would not (cooling centers specifically, heat index as a scenario variable) stayed in the extension.

This process was repeated across three source documents rather than performed once. An initial pass against the CMDDS produced a working ontology. Cross-checking that ontology against the framework's founding paper [1] surfaced a structural omission: the CMDDS describes translation *stages* in detail but never formalizes PAPO's own stated four-component *architecture*, because that architecture is defined in the founding paper, not the CMDDS. A third pass against the textbook chapter presenting PAPO in full [3] confirmed the core/extension boundary independently — its own two-part structure (a generic model section, then a heatwave-specific application section) mirrors the one we had already drawn — and surfaced two further gaps: an individual-level exposure signal the CMDDS had not distinguished from scenario-level environmental data, and a resource-level attribute (air-conditioned vehicles) distinct from the household-level attribute the CMDDS already captured.

We highlight this not to claim the ontology is now complete, but because the discrepancies found across passes are, we think, the actual value of doing this work formally: they are the kind of gap a prose reading tends to smooth over without noticing, and a structured one tends to expose.

4. The PAPO-Core Ontology, in Plain Language

The core ontology organizes PAPO's concepts into fifteen top-level categories, each mutually exclusive by design (Table 1 lists the concepts most relevant to a general reader; the full formal schema is in the accompanying repository).

Table 1. Selected PAPO-Core concepts.

Concept	Plain-language meaning
Policy	Institutional rules, thresholds, and priorities governing response to a hazard
Opportunity	A concrete, personalized, feasible action generated for an individual or segment — may be null
PopulationSegment	A group sharing vulnerability, access, or exposure characteristics

Concept	Plain-language meaning
Resource	An exhaustible, location-aware asset (facility capacity, vehicles, volunteers)
RiskProfile	An individual or segment's combined baseline vulnerability and current exposure
SystemComponent	One of PAPO's four architectural modules (see Figure 1)
PrioritizationRule / EquityMetric	Governance-approved rules for allocating scarce resources fairly

Figure 1 renders the core translation logic as a flowchart. It is the single most useful entry point into the framework: a trigger activates a policy; the Personalized Opportunity Engine checks whether a feasible resource exists for a given population segment's risk profile; if yes, an opportunity is delivered through an appropriate channel; if no, the system produces a null output and escalates through one of five recognized failure pathways rather than silently failing. Either way, results feed back into the Policy Input Layer — on a fast (tactical) or slow (strategic) timescale, kept deliberately separate because they involve different decision-makers and different time horizons.

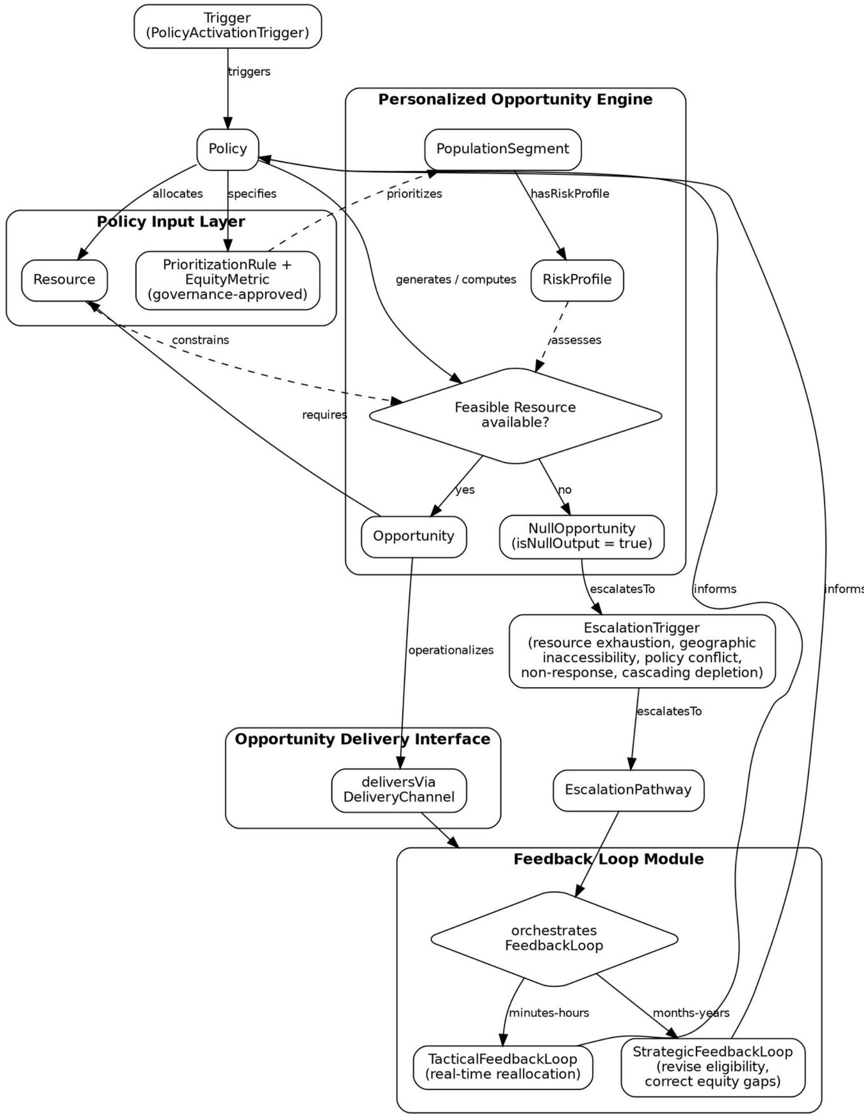

Figure 1. PAPO-core translation logic.

Solid arrows are the primary sequence; dashed arrows are supporting relationships that hold throughout rather than being a single step. The two diamonds are the framework's two required decision points: whether a feasible resource exists, and which feedback timescale applies.

5. Applying the Core to Heatwaves

The heatwave extension adds only what does not generalize (Table 2). Everything else — the five failure pathways, the governance requirements, the population-segment logic — is inherited unchanged from the core.

Table 2. Core versus heatwave-specific, illustrative examples.

Layer	Examples
Core (hazard-agnostic)	Escalation pathways for resource exhaustion, geographic inaccessibility, policy conflict, non-response, cascading depletion; tactical/strategic feedback separation; governance-approved equity metrics; outdoor-occupation and unhoused population segments
Heatwave extension	Heat-advisory triggers; cooling-center resources; ambient temperature, heat index, and UV index as scenario inputs; indoor temperature as an individual-level exposure signal, distinct from outdoor scenario conditions; wellness checks as an escalation pathway

This split is not merely tidy. It is a template: a wildfire-smoke, flood, or cold-snap application would extend the same core, contribute its own delta, and inherit the governance and failure-mode logic without re-deriving it. That reuse is the point of formalizing the core separately in the first place.

6. Visualization as a Bridge

We want to name explicitly what we think is doing the actual work of lowering the barrier to reuse: not the ontology, but Figure 1. The ontology gives the diagram its precision — every arrow in Figure 1 corresponds to a specific, checkable relationship in the formal schema, not an informal gloss. But the diagram is what a new reader will actually look at first, and in our experience, what most readers stop at. A specification a reader has to work through to understand is a specification most readers will not finish. A specification a reader can see in one figure, and then verify against prose and formal detail only where they want to go deeper, has a much better chance of being used.

This is why the formal ontology and the diagram are published together but serve different readers: the diagram for anyone deciding whether the framework is relevant to their work, the ontology for anyone who decides to build on it.

7. Discussion

Generalizability. The core/extension split was validated three times over — the founding paper's own generalizability claim, the textbook's parallel generic/heatwave structure, and the process of building the ontology itself, which repeatedly found that content assumed to be heatwave-specific was not. This gives us reasonable confidence the core will hold for other hazards, though it has not yet been tested against one.

Governance. A machine-checkable schema is a natural companion to governance review processes like the MOKG-PH framework the CMDDS already operates under: a reviewer can ask whether a proposed deployment's policy object actually specifies an approved equity metric, rather than trusting a narrative claim that it does.

Limitations. This is schema-level work only (a TBox, in ontology terms): no instances, no implemented knowledge graph, no field deployment. It formalizes what PAPO claims, not whether those claims hold up in practice — that question is separate, and, as we have argued elsewhere, only partially answered by historical evidence for PAPO's underlying mechanism [4]. Whether formalization actually increases uptake, as opposed to merely making the framework easier to evaluate, is itself an open, testable question this paper cannot answer in advance.

Future work. Three extensions follow directly from this one. First, populating the ontology with actual instance data (a knowledge graph, in a technology such as Neo4j or an RDF triplestore) for a specific city would be the natural next layer, though we deliberately have not built one yet — doing so before a concrete use case exists risks encoding one contributor's implementation choices as if they were part of the theory. Second, applying the same core ontology to a second hazard (wildfire smoke is the most immediate candidate, given comparable timing and resource-allocation structure) would be a direct test of the generalizability claim above, rather than an inference from document cross-checking alone. Third, and most directly testable, is the question this paper exists to address: whether a diagram-led, plain-language entry point measurably changes engagement with the underlying repository.

8. Conclusion

We formalized PAPO as a two-layer ontology and rendered its core logic as a single flowchart, not to replace the CMDDS but to give it a faster entry point. If the six-month absence of forks reflected a legibility barrier rather than a lack of interest, this should measurably help; if it does not, that itself is useful information about what the framework actually needs. Both artifacts are open for reuse, critique, and — we hope — forking.

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Declaration of Competing Interests

The author declares no competing interests.

Data and Code Availability

Both artifacts described in this paper are openly available in the `knowledge-engineering/` directory of the framework's GitHub repository: **https://github.com/Min-PH/PAPO-Heatwave-AI**

- `papo_core_ontology.yaml` — the hazard-agnostic core ontology (Section 4)
- `papo_heatwave_extension_ontology.yaml` — the heatwave-specific extension (Section 5)

- `papo_core_flowchart.md` — the flowchart source reproduced as Figure 1

The repository also hosts the CMDDS specification [2] and a companion evidence review for PAPO's underlying mechanism [4]. All files are versioned and changelogged; each ontology file documents, in its own header, which source document motivated each addition.

AI Disclosure Statement

This paper's prose, and the accompanying formal ontology and flowchart it describes, were produced through iterative collaboration between the author and Claude (Anthropic), a large language model AI system. Specifically: the AI system performed the knowledge-engineering work of translating the PAPO framework's prior publications [1–3] into a formal, two-layer, OWL-style ontology and an accompanying flowchart, under the author's direction and with the author reviewing, correcting, and approving each modeling decision (including class boundaries, disjointness axioms, and the core/extension split) across multiple working sessions. The AI system also drafted this paper's prose from an outline and scope set by the author, which the author subsequently revised in structure, framing, and content. The author is responsible for, and has reviewed, all claims, citations, and conclusions in this paper and the accompanying repository files. No AI system is credited as an author of this work.

References

[1] Wu, Min. *PAPO: A Policy-Aware Personalized Opportunity Framework for Heatwave Public Health Interventions.* February 8, 2026. doi:10.2139/ssrn.6198260

[2] Wu, M. *CMDDS — PAPO Heatwave Framework, v1.2.* Zenodo, 2026. DOI: 10.5281/zenodo.18637394

[3] Wu, M. *Artificial Intelligence in Public Health.* Springer Nature, 2026. Section 8.7.

[4] Wu, M. *Mechanism Evidence for PAPO's Opportunity-vs-Alert Mechanism.* MECHANISM_EVIDENCE.md, 2026. Available at: https://github.com/Min-PH/PAPO-Heatwave-AI